

Ehsan Ghoreishi

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EDUCATION

Virginia Tech

Ph.D. in Computer Engineering, Total GPA: 3.80/4

Blacksburg, VA, U.S.

2024-present

Virginia Tech

M.Sc. in Computer Engineering, Total GPA: 3.77/4

Blacksburg, VA, U.S.

2024

Isfahan University of Technology

B.Sc. in Electrical Engineering, Total GPA: 3.72/4

Isfahan, Iran

2021

RESEARCH EXPERIENCE

Real-Time GPU Acceleration for URLLC/eMBB Multiplexing

Complex Network & Security Research, Virginia Tech

2022-present

Deep Reinforcement Learning Solution for URLLC/eMBB Multiplexing

Complex Network & Security Research, Virginia Tech

2022-2024

Complex-Valued Neural Network for Wireless Modulation Classification

Isfahan University of Technology Signal Processing Research Center

2020-2021

Designed and Analyzed Nonlinear Effects in Remote Wire Monitoring.

Isfahan University of Technology Optic Research Center

2020-2021

Calculation of Heat Distribution of Double Brillouin Pulse in Water

Isfahan University of Technology

2021

WORK EXPERIENCES

Graduate Research Assistant, CNSR Lab, Virginia Tech, Blacksburg, VA

Graduate Teaching Assistant, Bradley Dept. of Electrical and Computer Engineering, Virginia Tech, Blacksburg, VA

Research Assistant, Electrical and Computer Department, Isfahan University of Technology, Isfahan, Iran.

Research Assistant, Optical and Signal Processing Research Center, Isfahan University of Technology, Isfahan, Iran.

PUBLICATIONS

Papers.....

Ehsan Ghoreishi, Bahman Abolhassani, Yan Huang, Shiva Acharya, Wenjing Lou, Y. Thomas Hou "Cyrus: A DRL-based Puncturing Solution to URLLC/eMBB Multiplexing in O-RAN." in 2024 33rd International Conference on Computer Communications and Networks (ICCCN), 2024, pp. 1–9. (Invited Paper)

Ehsan Ghoreishi, Bahman Abolhassani, Yan Huang, Shiva Acharya, Wenjing Lou, Y. Thomas Hou "Cyrus+: A Real-Time DRL-based Puncturing Solution to URLLC/eMBB Multiplexing in O-RAN.", (Submitted to *IEEE Transactions on Machine Learning in Communications and Networking*)

Ehsan Ghoreishi, Shiva Acharya, Wenjing Lou, Y. Thomas Hou "Enhancing 5G Efficiency: How Machine Learning is Revolutionizing eMBB and URLLC Puncturing.", (To be submitted to *IEEE Wireless Communica-tions*)

Poster.....
Ehsan Ghoreishi, Bahman Abolhassani, Yan Huang, Shiva Acharya, Wenjing Lou, Y. Thomas Hou "Cyrus: A DRL-based Puncturing Solution to URLLC/eMBB Multiplexing in O-RAN." Office of Naval Research (ONR) MURI Meeting, 2024.

AWARDS AND ACHIEVEMENTS

CCI SWVA Cyber Innovation Scholar Recipient, Commonwealth Cyber Initiative – SWVA (Funding Number: 556194)	2025
Nationwide University Entrance Exam – Mathematical Physics Ranked 90 th among 200,000 participants nationwide	2017

TEACHING EXPERIENCE

The Graduate Teaching Assistant in Introduction to Computer Networking Bradley Dept. of Electrical and Computer Engineering, Virginia Tech	2022
The Teaching Assistant in Principles of Electronics Electrical and Computer Engineering Dept., Isfahan University of Technology	2021
The Teaching Assistant in Electric Circuits II Electrical and Computer Engineering Dept., Isfahan University of Technology	2021
Course Instructor in MATLAB Electrical and Computer Engineering Dept., Isfahan University of Technology	2022

RESEARCH SUPPORT

ONR/Office of Naval Research
Principal Investigator: Jeffery H. Reed
NSF/National Science Foundation
Principal Investigator: Thomas Hou
CCI/Virginia Commonwealth Cyber Initiative
Principal Investigator: Yi Shi
ICTAS/Virginia Tech Institute for Critical Technology and Applied Science
Principal Investigator: Thomas Hou

SKILLS

Programming: Python, MATLAB, C/C++, Assembly
Experienced with numerical and simulation tools: TensorFlow, TF-Agents, PyTorch, MATLAB, Anaconda, CST, Maxwell, MATLAB Simulink

PROFESSIONAL MEMBERSHIPS

Institute of Electrical and Electronics Engineers (IEEE)	2022-present
IEEE Communications Society (ComSoc)	2022-present